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To the Editors of PLOS ONE,

I am submitting our manuscript, "Benchmarking 1,982 classifier configurations for early severity prediction in acute pancreatitis: an exploratory multi-dataset analysis," for consideration as a Research Article in PLOS ONE.

Severe acute pancreatitis (SAP) carries substantial morbidity and mortality, and most prior machine-learning studies of SAP severity prediction compare only a small number of algorithm families, rarely with formal significance testing against a simple baseline. This manuscript reports, to our knowledge, an unusually large systematic comparison — 1,982 classifier configurations spanning linear models, tree ensembles, six independent gradient-boosting implementations, two additional boosting-algorithm variants, deep learning architectures, and a hybrid ensemble — evaluated under one identical cross-validation protocol on two independent public de-identified datasets, together with Hanley-McNeil significance testing against a logistic-regression baseline and an engineered lab-only organ-dysfunction feature.

We believe this work is well suited to PLOS ONE because its primary contribution is methodological rigor and transparent reporting of both positive and negative/mixed findings (the top-ranked model did not reach conventional

statistical significance on one dataset; an engineered clinical feature did not consistently improve performance), rather than a single striking discrimination result. We report these findings honestly rather than emphasizing only the more favorable results.

This manuscript is not under consideration elsewhere. All authors have approved the manuscript and this submission. We have no competing interests to declare. This study is a secondary analysis of two previously published, de-identified public datasets; no new human-subject data were collected. Please see the enclosed Ethics Statement, which describes this situation directly: the original data-collection ethics oversight was reported by the original dataset authors and was not independently re-verified beyond the datasets' public licensing and de-identification status. We are glad to provide any further documentation the editors or reviewers require.

All code, model configurations, benchmark results, and from-scratch reference implementations described in the manuscript are openly available (see Data Availability Statement).

Thank you for considering our manuscript.

Sincerely,

Netanel Stern

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